

Final

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Clean / prepare data for relevant modeling

```
# Load packages
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.3
```

```
## Warning: package 'readr' was built under R version 4.5.3
```

```
## Warning: package 'forcats' was built under R version 4.5.3
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.0      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.2      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(caret)
```

```
## Loading required package: lattice
```

```
##
```

```
## Attaching package: 'caret'
```

```
##
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
## lift
```

```
library(cluster)
```

```
## Warning: package 'cluster' was built under R version 4.5.3
```

```
set.seed(4827)
```

```
# Load dataset
```

```
fuel <- read_csv("monthly_fuel_costs.csv")
```

```
## Rows: 296604 Columns: 21
## -- Column specification -----
## Delimiter: ",",
## chr  (11): plant_name_eia, utility_name_eia, state, fuel_type_code_pudl, fue...
## dbl  (9): plant_id_eia, plant_id_pudl, utility_id_eia, utility_id_pudl, fue...
## dtm   (1): report_date
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
# Data overview
#glimpse(fuel)

# Null percentages
sapply(
  fuel[c("mercury_content_ppm",
         "moisture_content_pct",
         "chlorine_content_ppm")],
  function(x) mean(x == "null" | is.na(x)) * 100
)

# Uniqueness testing
sapply(fuel, function(x) length(unique(x)))

#table(fuel$data_maturity)
#table(fuel$fuel_cost_per_mmbtu_source)
#table(fuel$fuel_type_code_pudl)

# Data cleaning and selecting relevant fields for fuel quality modeling
fuel_clean <- fuel %>%
  select(
    fuel_mmbtu_per_unit,
    sulfur_content_pct,
    ash_content_pct,
    mercury_content_ppm
  ) %>%

  # convert "null" strings to NA
  mutate(
    mercury_content_ppm = na_if(mercury_content_ppm, "null"),
    mercury_content_ppm = as.numeric(mercury_content_ppm)
  )

#glimpse(fuel_clean)
```

Set sampling and clustering

```
fuel_sample <- fuel_clean %>%
  sample_frac(0.02)

# Median imputation
median_mercury <- median(fuel_sample$mercury_content_ppm, na.rm = TRUE)

cluster_vars_q <- fuel_sample %>%
```

```

mutate(
  mercury_content_ppm = ifelse(
    is.na(mercury_content_ppm),
    median_mercury,
    mercury_content_ppm
  )
) %>%
drop_na()

# Scale variables for normalization
preProc <- preProcess(cluster_vars_q, method = c("center", "scale"))
cluster_scaled_q <- predict(preProc, cluster_vars_q)

# Cluster output
# glimpse(cluster_scaled_q)

```

Run elbow & silhouette methods to view optimal K

Fit the model and summarize

```

# Elbow
wss <- map_dbl(1:10, function(k) {
  kmeans(cluster_scaled_q, centers = k, nstart = 20)$tot.withinss
})

elbow_df <- tibble(
  k = 1:10,
  wss = wss
)

# Silhouette
sil_width <- map_dbl(2:10, function(k) {
  km <- kmeans(cluster_scaled_q, centers = k, nstart = 20)

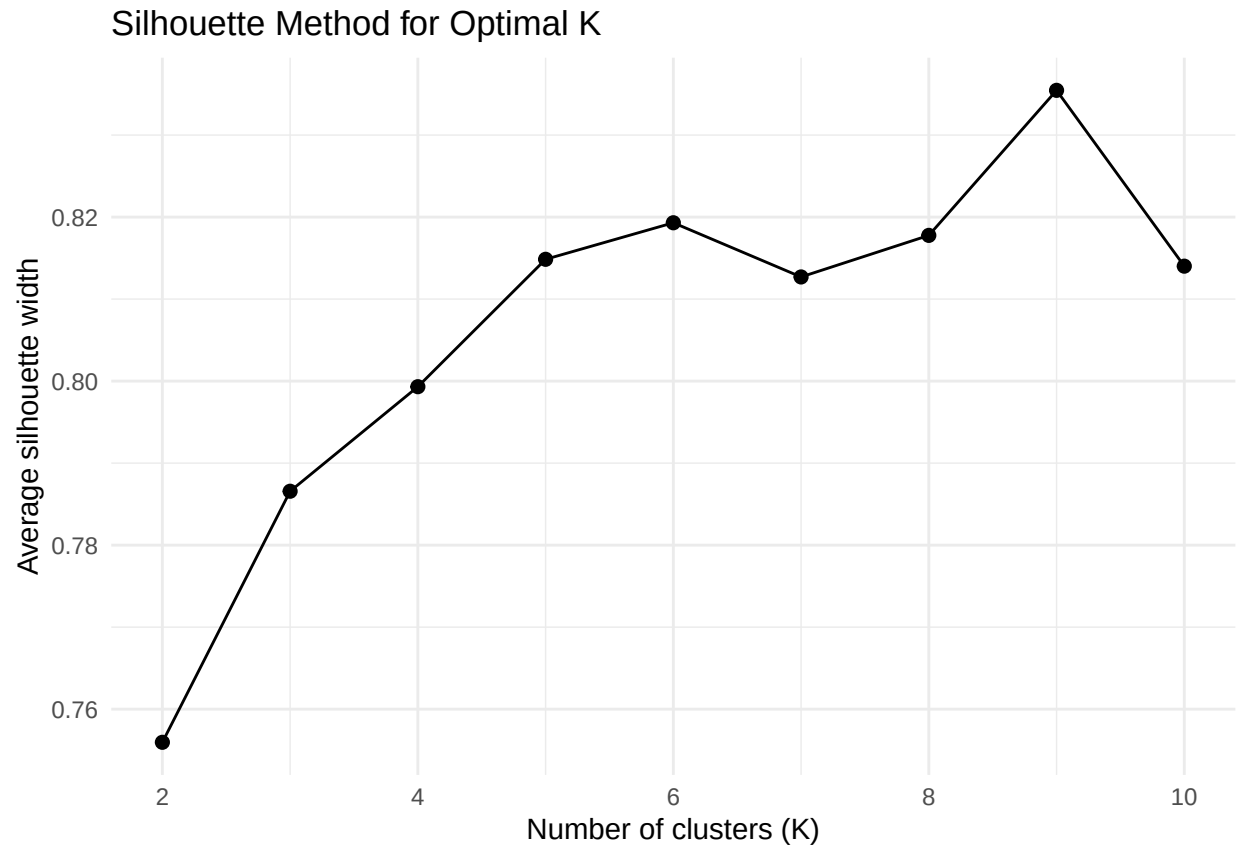
  ss <- silhouette(km$cluster, dist(cluster_scaled_q))

  mean(ss[, 3])
})

sil_df <- tibble(
  k = 2:10,
  silhouette = sil_width
)

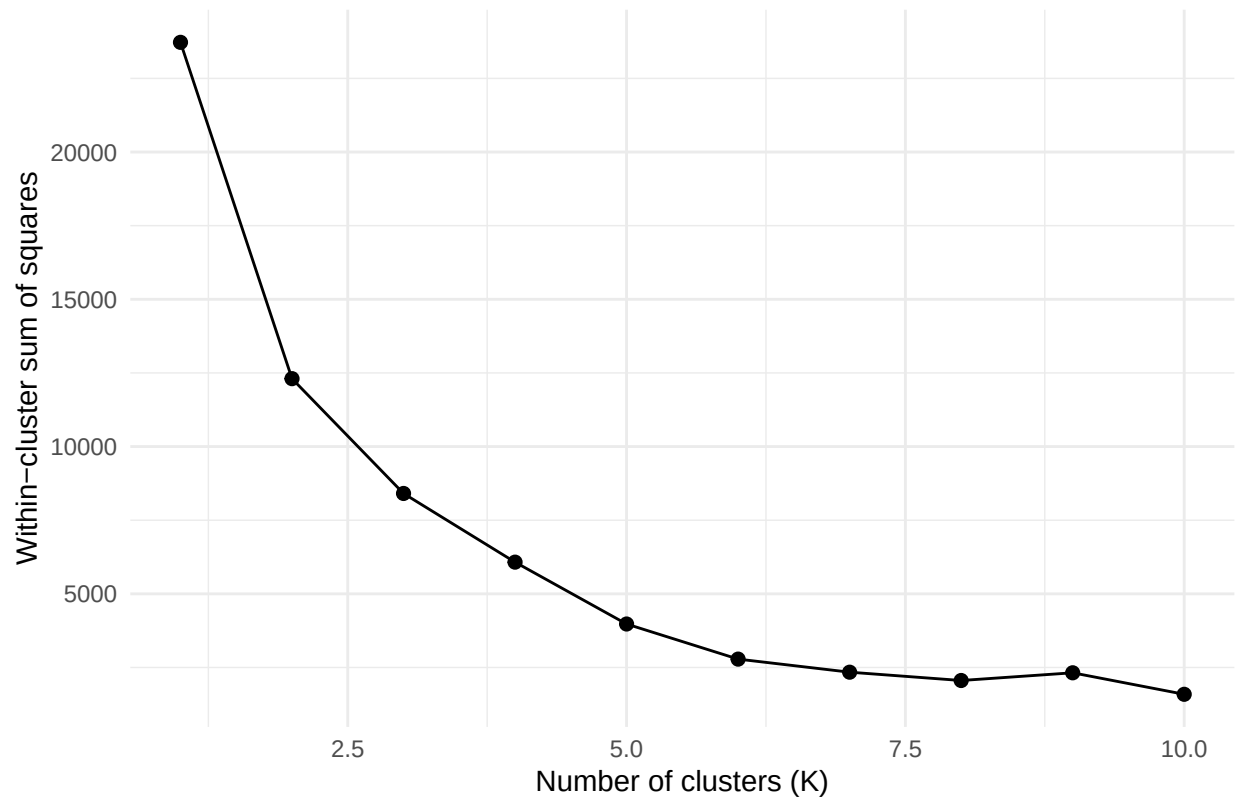
ggplot(sil_df, aes(k, silhouette)) +
  geom_line() +
  geom_point(size = 2) +
  theme_minimal() +
  labs(
    title = "Silhouette Method for Optimal K",
    x = "Number of clusters (K)",
    y = "Average silhouette width"
  )

```



```
# Plot elbow curve
ggplot(elbow_df, aes(x = k, y = wss)) +
  geom_line() +
  geom_point(size = 2) +
  theme_minimal() +
  labs(
    title = "Elbow Method for Fuel Quality Clustering",
    x = "Number of clusters (K)",
    y = "Within-cluster sum of squares"
  )
)
```

Elbow Method for Fuel Quality Clustering



```
k_final <- 4

final_kmeans <- kmeans(cluster_scaled_q, centers = k_final, nstart = 20)

cluster_results <- cluster_vars_q %>%
  mutate(cluster = final_kmeans$cluster)

# -----
# 8. Cluster summaries
# -----

cluster_results %>%
  group_by(cluster) %>%
  summarise(across(everything(), mean), .groups = "drop")
```

The elbow method would argue that $K=4$ is where results start to truly produce diminishing returns. Although the elbow method appears like reasonably $K=4$, the Silhouette analysis indicates that $K=5$ has a more optimized width before sharply declining at $K=6$, making $K=5$ a more stable and real clustering with the highest cohesion and separation. However, $K=4$ will be a little more stable with less extreme outliers. Less suggests slight under-resolved structures and more looks like over-segmentation.

View detailed profile Centroids

```
km4 <- kmeans(cluster_scaled_q, centers = 4, nstart = 20)

clustered_data <- cluster_vars_q %>%
  mutate(cluster = km4$cluster)
```

```

cluster_profile <- clustered_data %>%
  group_by(cluster) %>%
  summarise(
    n = n(),
    across(
      where(is.numeric),
      list(mean = mean, sd = sd),
      .names = "{col}_{fn}"
    ),
    .groups = "drop"
  )

cluster_profile

km3 <- kmeans(cluster_scaled_q, centers = 3, nstart = 20)

table(
  K3 = km3$cluster,
  K4 = km4$cluster
)

```

Looking at the centroid group's qualities, under K=4, cluster 3 is assuming 77% of the observations and extremely low mean mmbtu. Cluster 2 and 4 are almost identical, with the key difference being their sulfur content seems to define their fuel quality grouping rather than a new fuel type, while Cluster 1 seems to be uniquely off set. The key take-away from these discoveries is that subgroup characterizations show prevalent changes that the silhouette increase alone does not distinguish.

Going from K=3 to K=4 shown in the cross-section reveals the cluster is defined by environmental distinction rather than random assignment via high mercury content. K=5 may split an existing cluster and increase the silhouette slightly, but one cluster will be divided into two similar clusters. K=4 feels like a reasonable mid-point that yields high sulfur/ash (coal), mid-grade coal, gas-like fuel, and mercury-heavy coal. Having full sets of data might prove more useful clusters for future evaluation, but there were many null values to work around mercury, moisture, and chlorine.